

Richie Tan

Los Gatos, CA | 669-291-6042 | richietan2004@gmail.com | richietan.dev | linkedin.com/in/richie-tan | github.com/richtan

EDUCATION

Purdue University

West Lafayette, IN

Bachelor of Science in Computer Science, GPA: 3.73/4.0

August 2023 – December 2026

Coursework: Data Structures, Algorithms, Systems Programming, Operating Systems, Computer Networks

EXPERIENCE

Software Engineering Intern

June 2026 – August 2026

Capital One

Richmond, VA

- Built Connex Autofix, a Claude tool that turns Jira tickets into reviewable Lit/TypeScript PRs
- Saved 360+ developer hours and \$21,900 across a 116-ticket backlog, with recurring savings on new tickets
- Drove design-team adoption with live demos, hands-on install support, and fast issue turnaround
- Reduced component review from 10 minutes to under 30 seconds with a before-and-after render pipeline

Software Engineering Intern

May 2025 – August 2025

Pacific Northwest National Laboratory

Richland, WA

- Reduced SQL query latency from 10 seconds to sub-second by replacing JOINS with an indexed cache table
- Built a RAG-powered Semantic Kernel chat agent in C# enabling natural-language access to internal APIs
- Simplified the contacts database API and optimized database schema to support the new chat agent backend

Founding Software Engineer

August 2024 – Present

AlphaSeeker AI

Boston, MA (Remote)

- Built a Next.js financial insights platform end-to-end, from initial architecture to production deployment
- Implemented an automated error-tracking system with email notifications and detailed, traceable logging
- Automated daily price forecasting for 26,000 stock tickers with nearly 60% directional accuracy
- Reduced search load times from 20 seconds to near-instant by optimizing PostgreSQL queries and adding indexes
- Led a team of 3 developers to build forecast visualization interfaces, enabling price prediction comparisons

Scientific Applications Assistant

May 2024 – Present

Purdue's Rosen Center for Advanced Computing (RCAC)

West Lafayette, IN

- Shipped a redesigned web dashboard across 4 HPC clusters, serving 12,000+ users
- Presented research on HPC dashboard design at SC25 and GOOD 2026 conferences
- Reduced load times by 50% by implementing Ruby on Rails backend caching and IndexedDB frontend caching
- Built job data visualizations with Chart.js and Slurm data, enabling users to analyze 100,000+ job records
- Open-sourced the dashboard on GitHub, attracting adoption interest from multiple HPC institutions

PROJECTS

Embedded OS Kernel (XINU) | C, ARM

- Added message-passing system calls to the XINU kernel, cross-compiled for ARM on BeagleBone Black
- Built bounded-buffer producer/consumer synchronization with semaphores and interrupt masking

XENONnT Dark Matter Research | Python, NumPy, SciPy | 3rd Place, Purdue Research Symposium

- Tagged Rn-Po decay pairs in XENONnT event data and subtracted accidental coincidences per spatial bin
- Filtered spatial bins by pair purity and mapped radon-daughter drift velocity fields with numba

PrePrompt | TypeScript, Node.js, Turborepo

- Published an open-source CLI tool on npm that benchmarks prompts on AI coding agents in parallel sandboxes
- Automated verifiable success criteria generation from prompts and agent output evaluation

TECHNICAL SKILLS

Languages: C++, C, Python, C#, TypeScript, JavaScript, SQL, Java, Bash, Scala

Frameworks/Databases: Next.js, Ruby on Rails, FastAPI, PostgreSQL, Redis, Tailwind CSS

Developer Tools: Git, Linux, GCC, GDB, GitHub Actions, Docker, GCP, Slurm

Libraries: React, pandas, NumPy, matplotlib, polars, Chart.js